

## SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

### ASSESSMENT TOOL 2 – Concept Mapping

|                  |                                                                                                                         |               |                                     |
|------------------|-------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------|
| Course Code:     | CSA03                                                                                                                   | Course Title: | Data Structures                     |
| CO Assessed:     | CO1 – Imitation (BL3, BL4)                                                                                              | Assessment:   | Assessment Tool 2 – Concept Mapping |
| Weightage:       | 10% of CIA                                                                                                              | Total Marks:  | 50 (5 Questions × 10 Mark)          |
| CO5 Statement:   | Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity (BL3, BL4) |               |                                     |
| Student Name:    | T.VAISHNAVI                                                                                                             | Reg. No.:     | 192421370                           |
| Section / Batch: | B-TECH(IT)                                                                                                              | Date:         | 17/07/2026                          |

#### Code of Conduct

I (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

#### Instructions

- This test contains 5 Application-Based Questions Total: 50 Marks.
- Evaluation of Answer Quality - Concept Identification, Linkages, Organization, Integration of Literature and clarity
- No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

| Section / Topic                                                | Focus                                 | Q Nos   | Marks |
|----------------------------------------------------------------|---------------------------------------|---------|-------|
| Introduction to Data Structure- Abstract Data Types (ADT)      | ADT                                   | 1,2     | 20    |
| Types of Data Structure- Linear and Non linear Data structures | Linear and Non linear Data structures | 3,4     | 20    |
| Recursion , Performance Analysis and Measurement               | Performance Analysis                  | 5       | 10    |
| GRAND TOTAL:                                                   |                                       | 50Marks |       |

## Annexure A – Concept Map Questions

Q1. A system needs to search for specific records in a large dataset efficiently. Construct a concept map linking Searching Algorithm → Linear Search → Binary Search → Time Complexity → Efficiency. Include relationships that compare the number of operations and performance of each method. Analyze which algorithm is more efficient and justify your conclusion.

Q2. A library management system stores thousands of book records including title, author, and availability status. Construct a concept map linking Data → Data Structure → Organization → Access Methods → Efficiency. Include different data structures used for storing and retrieving records. Explain how proper organization improves search efficiency and system performance

Q3. A weather monitoring system collects temperature, humidity, and wind data continuously. Construct a concept map linking Primitive Data Types → Non-Primitive Structures → Linear Structures → Non-Linear Structures → Applications. Include examples and explain how each structure supports continuous data processing..

Q4. A logistics company processes delivery requests across multiple cities. Construct a concept map linking Algorithm → Input Size → Time Complexity → Space Complexity → Scalability → Efficiency. Analyze how algorithm selection impacts system scalability and justify the best approach for handling large datasets.

Q5. A printer system manages print jobs submitted by multiple users. Construct a concept map connecting Queue → FIFO → Job Scheduling → Processing → Efficiency and Stack → LIFO → Undo Operations → Execution. Explain how each structure is applied and justify their suitability in this context.

### Rubrics for Calculation

| Criteria                  | Weightage | Level 4 – Exemplary                                                         | Level 3 – Proficient                              | Level 2 – Developing                                        | Level 1 – Beginning                     |
|---------------------------|-----------|-----------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------|-----------------------------------------|
| Concept Identification    | 25%       | Identifies all key concepts from literature accurately and comprehensively  | Identifies most key concepts with minor omissions | Identifies limited concepts; some are irrelevant or missing | Fails to identify essential concepts    |
| Linkages                  | 25%       | Establishes clear, meaningful, and accurate relationships between concepts  | Shows correct relationships with minor gaps       | Relationships are weak, unclear, or partially incorrect     | No meaningful relationships established |
| Organization              | 20%       | Concepts are well-structured hierarchically with logical flow and grouping  | Mostly organized with minor structural issues     | Some organization but lacks clear hierarchy                 | No logical organization or structure    |
| Integration of Literature | 20%       | Integrates multiple studies effectively to show connections and comparisons | Shows some integration of literature              | Limited integration; mostly isolated concepts               | No integration of literature evidence   |

|         |     |                                                                        |                                               |                                                     |                                         |
|---------|-----|------------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------------|-----------------------------------------|
| Clarity | 10% | Concept map is clear, neat, visually structured, and easy to interpret | Generally clear with minor readability issues | Clarity is reduced; difficult to interpret in parts | Poorly presented and hard to understand |
|---------|-----|------------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------------|-----------------------------------------|

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
|                   |                    |                  |
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |



Name :- T. Vaishnavi

Reg :- 1921421370

Course code :- CSA0312 (Data Structure)

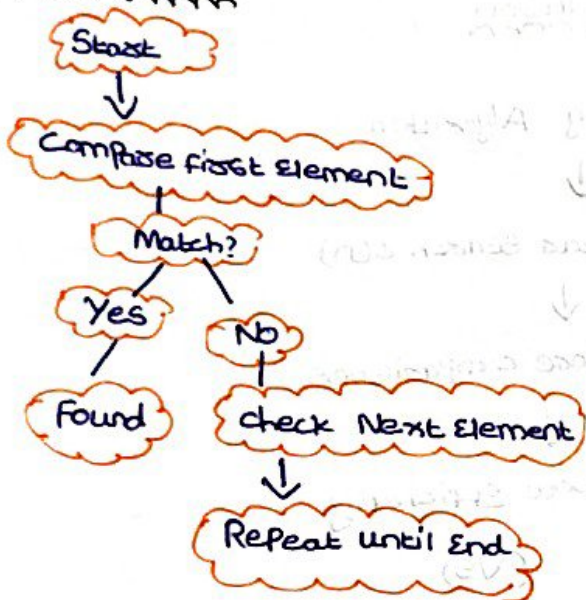
Assessment-2 (Concept Mapping)

## ① Searching Algorithms

• used to find a specific element in a collection of data

- Two common methods :- ① Linear search
- ② Binary search

### Linear Search



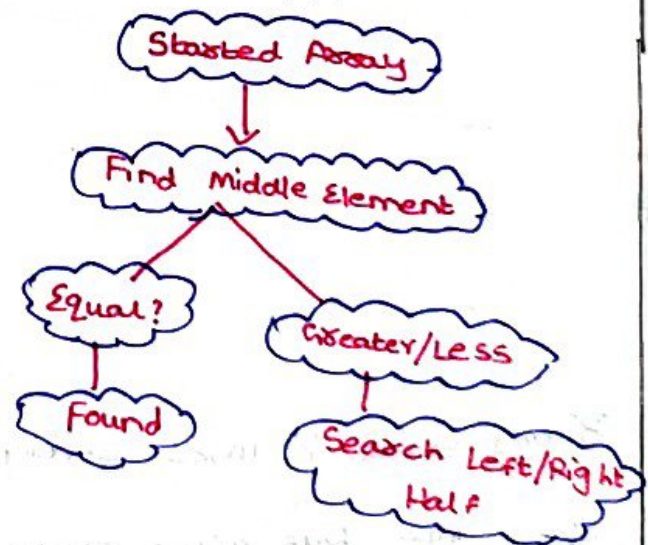
### features:-

- ✓ Sequential search
- ✓ work on sorted & unsorted data
- ✓ Easy to Implement
- ✓ Time complexity =  $O(n)$
- ✓ Space complexity =  $O(1)$

### features

- \* Requires sorted Data
- \* Faster searching
- \* Time complexity =  $O(\log n)$
- \* Space complexity =  $O(1)$

### Binary search



## Time complexity

Searching Algorithms

Time complexity

Linear  
 $O(n)$

Binary  
 $O(\log n)$

Small dataset

Linear search

Simple

Easy to use

## Conclusion

Searching Algorithm

Linear search  $O(n)$

More comparisons

Low Efficiency  
(Vs)

Binary search  $O(\log n)$

Less comparisons

Higher Efficiency

## Efficiency

Large Dataset

Binary search

Less comparisons

Faster Execution

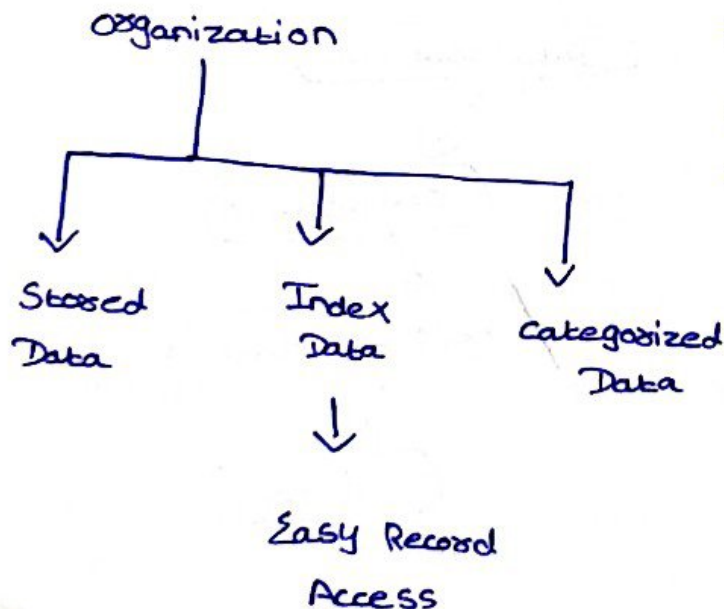
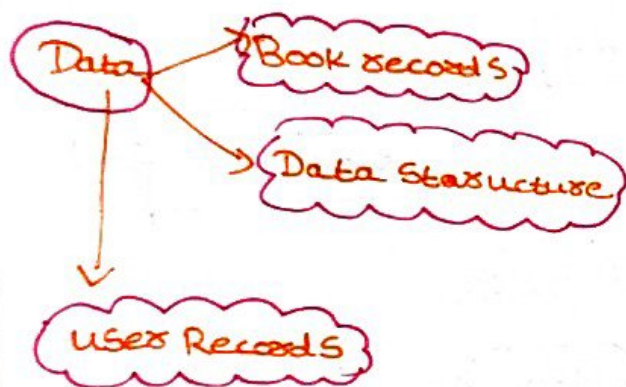
Binary search is more efficiency because it reduce the search space into half after every comparison, resulting in fewer operations and faster performance than Linear search.



# Library Management System

- Data is a collection of facts or information.
- In a library data includes:
  - Book Title
  - Author Name
  - Availability Status
  - Data stored using different structures for easy retrieval.

## Data organization



## Data Structure

- ① Array - Stores Records
- ② Linked List - Dynamic Storage
- ③ Hash Table - fast Searching

## Features

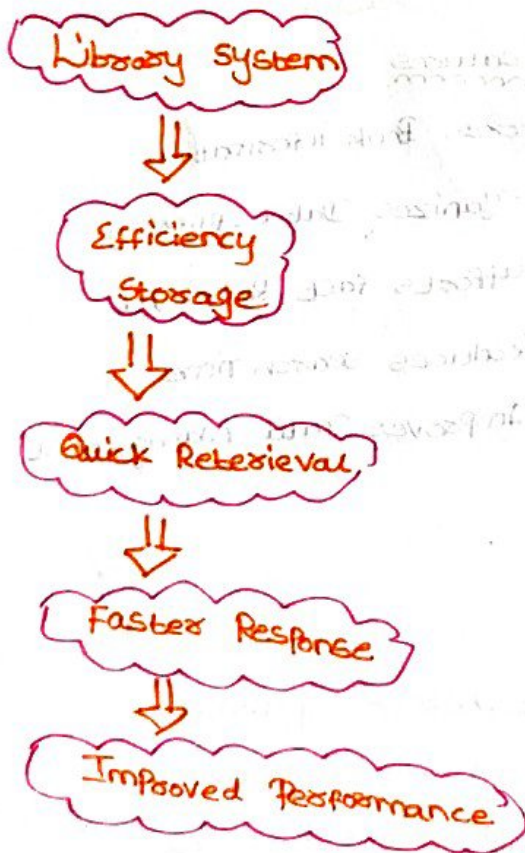
- \* Stores Book information
- \* Organizes Data efficiently
- \* Supports fast Retrieval
- \* Reduces search time
- \* Improves Data management

- ⇒ Sort by Title
- ⇒ Sort by Author
- ⇒ Index Book ID
- ⇒ Categorize by Subject
- ⇒ Reduces Duplicate Records

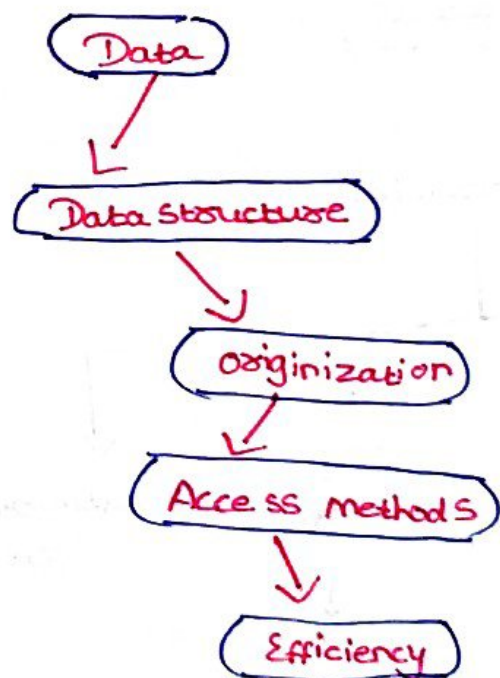
| <u>Access method</u> | <u>Purpose</u>                |
|----------------------|-------------------------------|
| Linear Search        | Search one record at a time   |
| Binary Search        | Search sorted records quickly |
| Hash Search          | Instant lookup using key      |
| Tree Search          | Hierarchical Searching        |



### System Performance



### Conclusion



Proper organization of library records using suitable data structure such as Arrays, Linked Lists, Hash Tables, and Trees improves search efficiency and system performance.

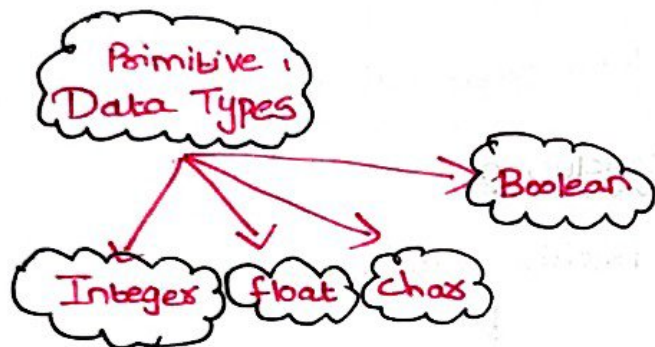


# Weather Monitoring System

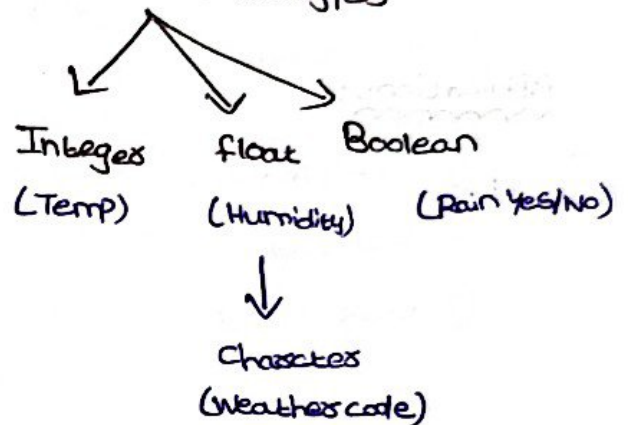
• A weather monitoring system continuously collects environmental data

• It records

- \* Temperature
- \* Humidity
- \* Wind Speed

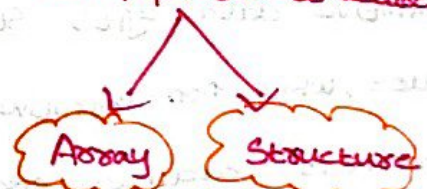


## Primitive Data types



- Stores single values
- Simple and fast
- Use less memory
- Easy to process
- Used for sensor readings

## Non Primitive Structures



- Stores multiple data values
- Organizes related information
- Easy data management
- Efficient processing

## Linear structure

## Application

Array

- Store hourly temperature

Queue

- continuous sensor data

Linked

- Dynamic weather records

List

## Recent data processing

Stack



## Non Linear Structures

- Graphs
- Tree

\* Hierarchical Storage

\* Fast Searching

\* Network connections

\* Efficient analysis

\* Better data organization.

### Applications:-

- ① Temperature
- ② Humidity
- ③ Wind Speed

### Conclusion

Primitive Data Types



Non-Primitive Data Types



Linear Structures



Non-Linear Structures



Applications



Continuous weather monitoring

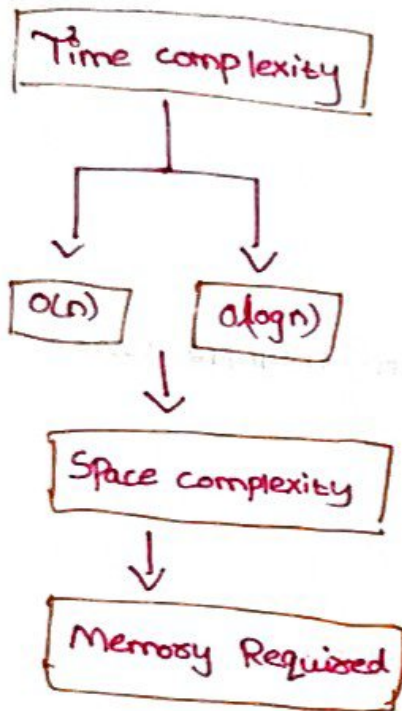
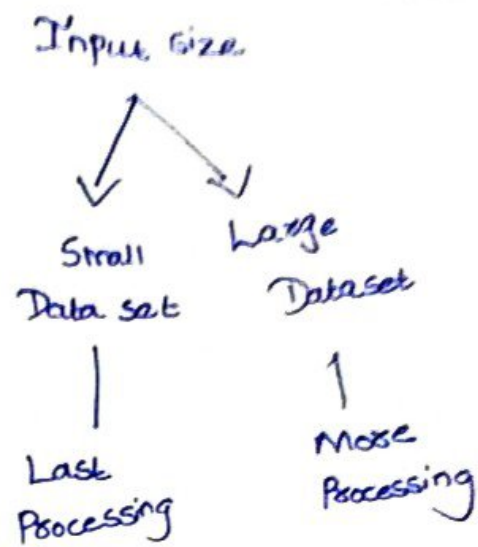
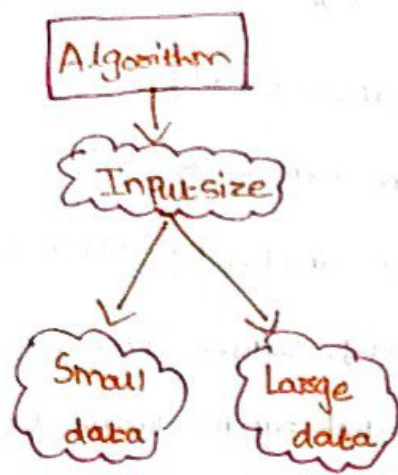
\* Primitive data types store weather values, while non-primitive, linear, and non-linear structures organize and process data.

## ④ Logistics company

### Algorithm

1) An algorithm is a step-by-step procedure used to solve a problem.

2) In a logistics company, algorithms process delivery requests across multiple cities.



- Number of delivery requests
- Number of cities
- Affects execution time
- Influence resource usage.

### Features

- 1) Time complexity measures execution time
- 2) Space complexity measures memory usage
- 3) Lower complexity gives better Performance
- 4) Suitable for large datasets

### Efficiency

Scalability



Handles more cities



More Fast Processing



High Efficiency



## Best Approach

Large Dataset  
↓  
Efficient Algorithm  
( $O(\log n)$ )

↓  
Less Time & memory

↓  
Better Performance

Efficient algorithms with low time and space complexity improve scalability, reduce processing time, and handle large delivery datasets efficiently.

## ⑤ Printer System

• A Printer System manages Print Jobs from multiple users.

It uses:-

- Queue for Print Job scheduling
- Stack for Undo operations
- These structures improve processing efficiency.

Printer system :- ① Queue  
② Stack

features

- first in first out (FIFO)
- Job processed in order
- Job is skipped
- Suitable for Print queues

Queue



FIFO



Job scheduling



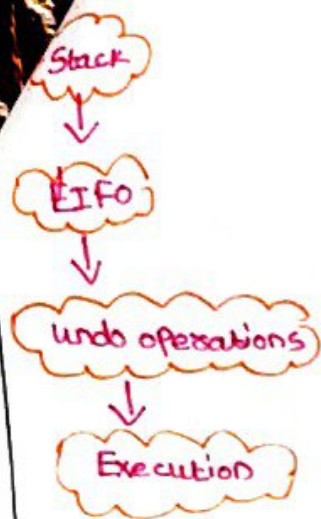
Print processing



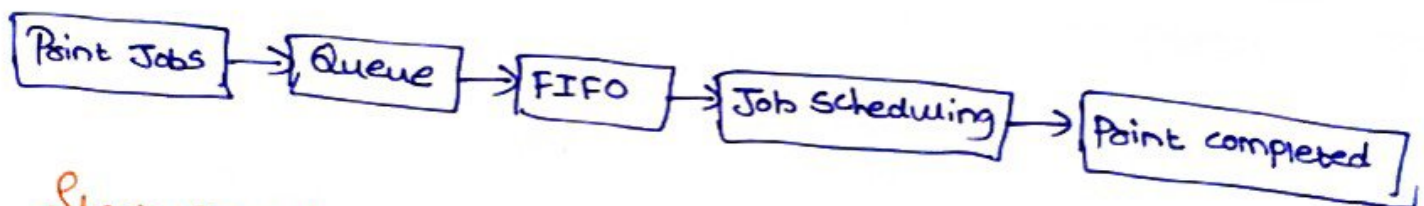
Efficiency

### Features

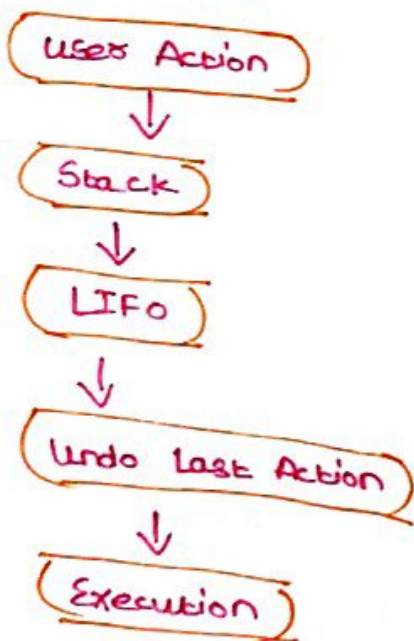
- Last in First out (LIFO)
- Stores recent actions
- Supports Undo ✓
- Fast execution
- Easy to implement



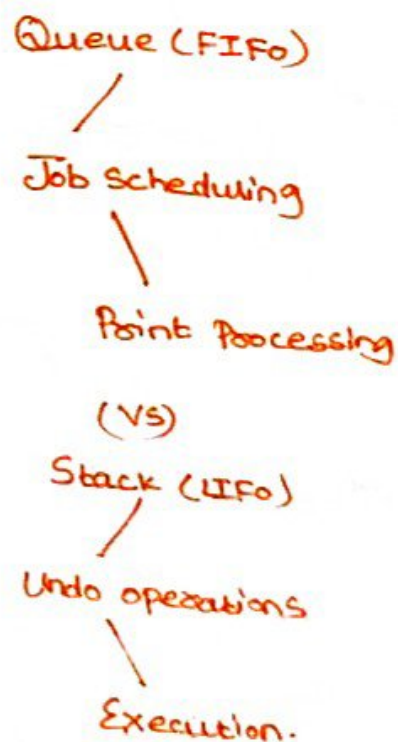
### Queue Processing



### Stack Execution



### Conclusion



Queue (FIFO) is best for Print Job Scheduling while Stack (LIFO) is ideal for undo operations. Together they improve Printer System efficiency and Performance.